

6G Weekly Survey — Jul 06 - Jul 12, 2026

15 items across 5 topics, ranked by importance.

6G Physical Layer

Multiuser MIMO-AFDM Beamforming for ISAC in Doubly Dispersive Channels

Rang Liu, Ming Li, A. Lee Swindlehurst, Norman Franchi, et al. · arXiv eess.SP · 2026-07-06

★★★★■ Strong contribution combining two critical 6G themes—AFDM waveforms for high mobility and ISAC beamforming—with rigorous signal processing and demonstrated superiority over OFDM in realistic scenarios. This paper addresses multiuser MIMO-ISAC in high-mobility doubly dispersive channels by formulating beamforming design for AFDM waveforms. A discrete affine Fourier transform (DAFT)-domain model captures Doppler-induced inter-bin coupling, and a joint WMMSE-MM algorithm optimizes sensing SNR and communication sum-rate under sector-illumination constraints. Simulations demonstrate AFDM outperforms OFDM in rate-sensing tradeoff and Doppler robustness.

Contribution: First joint beamforming optimization framework for multiuser MIMO-AFDM ISAC that explicitly handles Doppler coupling via DAFT-domain modeling and balances sensing quality with communication rate.

Relevance to your work: Directly aligns with your interests in AI-native air interfaces (alternative waveforms beyond OFDM), ISAC system design, and AFDM as a 6G candidate waveform for Doppler-resilient PHY.

6G Sensing Security: Distributed Game-Theoretic RL for Urban Beamforming and Attacker Detection

Parmida Geranmayeh, Onur Günlü · arXiv cs.IT · 2026-07-07

★★★★■ Addresses an important but under-explored security dimension of ISAC systems with a principled game-theoretic RL approach, though the evaluation appears limited to simulations without comparison to baseline security methods or real channel traces.

This paper addresses security in 6G ISAC systems by detecting active attackers that manipulate beamforming to increase interference in urban scenarios. The authors model the transmitter-attacker interaction as a game-theoretic problem and solve it using reinforcement learning to optimize beam allocation while detecting malicious manipulation. Simulations demonstrate the method can identify attackers attempting to redirect main beam lobes toward themselves instead of legitimate users.

Contribution: First application of game-theoretic RL to the specific problem of beamforming manipulation attacks in ISAC systems, where sensing capabilities are leveraged to detect adversarial behavior through channel observation.

Relevance to your work: Directly relevant to your ISAC research interests, particularly the intersection of beam management and sensing — the game-theoretic framework could inform robust beamforming strategies for AI-native air interfaces where adversarial resilience is critical for deployment.

6G Demos & Trials

Bending Beam for THz Wireless Networks: Fundamental, Design Issue, and Prototype

Aoran Liu, Xin Tong Hu, Weidong Mei, Ya Fei Wu, et al. · arXiv eess.SP · 2026-07-07

★★★★■ Addresses critical THz blockage problem with novel physical-layer approach backed by hardware validation, relevant to 6G sub-THz deployment challenges.

This paper reviews bending beams (non-diffracting, self-healing waveforms) for THz communications, addressing blockage bypass through curved propagation paths. It surveys beamforming design methods, hardware implementations, and application scenarios including communication and sensing. A THz hardware

prototype demonstrates performance advantages over conventional beam-focusing.

Contribution: First comprehensive review of bending beam technology for THz wireless with practical hardware demonstration showing blockage circumvention capabilities at THz frequencies.

Relevance to your work: Directly relevant to your 6G waveform and sub-THz research interests; bending beams offer alternative beam management strategy for blockage-prone THz channels and could inform ISAC applications requiring non-line-of-sight propagation.

Enabling Real-Time AI in O-RAN: Deploying and Measuring AI Inside a Near-RT RIC xApp

Lawrence Obiwewe, Krzysztof J. Rechowicz, Sampath Jayarathna, Safdar Hussain Bouk, et al. · arXiv cs.NI · 2026-07-02

★★★★■ Addresses a critical gap between theoretical AI/RAN integration and actual deployment by providing measured latency characterization and open tooling, directly informing practical O-RAN AI deployment strategies for 6G. This paper demonstrates the first measured deployment of AI inference directly within an O-RAN Near-RT RIC xApp on a live OpenAirInterface testbed, using logistic regression and shallow MLP models compiled as deterministic C code to classify RAN states from MAC/RLC/PDCP features. Measured inference latency is 1-25 microseconds with end-to-end service latency under 4 ms, meeting the 10 ms Near-RT budget in 95% of executions. The work validates the feasibility of embedding lightweight AI models in RIC control loops without external ML runtime dependencies.

Contribution: First practical demonstration of measured AI inference latency within a live O-RAN Near-RT RIC xApp, showing that deterministic C-compiled models can satisfy real-time constraints while the release of RIC Workbench enables reproducible research on commodity hardware.

Relevance to your work: Directly relevant as a reference architecture for deploying your AI-native PHY algorithms (neural receivers, learned channel estimation) within standardized O-RAN control loops, particularly for understanding real-time inference constraints in beyond-5G systems.

Repeated Contention Scheduling: A Novel Resource Allocation Algorithm Toward 6G Vehicular Networks

Alexey Rolich, Marco Tricco, Simone Paroli, Mert Yildiz, et al. · arXiv cs.NI · 2026-07-07

★★★★■ Solid incremental improvement to V2X resource allocation with hardware validation, but addresses a specific vehicular use case rather than broader 6G PHY innovation or AI-native architectures.

This paper introduces Repeated Contention Scheduling (RCS), a decentralized resource allocation algorithm for NR-V2X sidelink that replaces long-term reservations with multi-round feedback-driven contention. Simulations show RCS outperforms conventional Semi-Persistent Scheduling and Dynamic Scheduling in success probability, collision reduction, and timeliness metrics under high-density vehicular scenarios. An SDR-based testbed validates practical feasibility under realistic hardware impairments.

Contribution: The novel multi-round contention mechanism eliminates persistent reservation-based collisions in V2X while maintaining fully distributed operation, validated through both simulation and SDR hardware implementation.

Relevance to your work: Tangential—focuses on MAC-layer scheduling for vehicular networks rather than AI/ML PHY design, channel modeling, or the core technical areas in beyond-5G air interface research.

Agentic AI for IPoDWDM Network Lifecycle Automation: An MCP-Enabled Architecture

Chunmin Xia, Jakub Harbaczewski, Nikhil Dsilva, Julie Raulin, et al. · arXiv eess.SP · 2026-07-07

★★■ Competent network automation work but focused on optical transport layer management rather than air interface or wireless system innovations relevant to 6G.

The paper presents a multi-agent architecture using Model Context Protocol (MCP) for automating IP-over-DWDM optical transport networks, enabling end-to-end service lifecycle management and

closed-loop cross-layer control. The system integrates SDN principles with GNPpy optical modeling and telemetry to coordinate multi-vendor equipment in a vendor-agnostic manner. Experimental validation on an IPoDWDM testbed demonstrates practical feasibility of the distributed agent-based approach.

Contribution: First demonstration of MCP-based multi-agent architecture for cross-layer automation in multi-vendor optical transport networks with experimental validation.

Relevance to your work: tangential

Structural Analysis and Internal Stability Enhancement of Virtual-Admittance-Based Cascaded GFMs Under Unity Voltage-Feedback Decoupling

Stephen Arinze Obi, Ki-Hyun Kim, Jae-Jung Jung · arXiv eess.SY · 2026-07-07

★☆☆☆☆ This is power-electronics inverter control theory for grid integration, entirely outside the wireless physical-layer and air-interface research relevant to 6G systems.

This paper addresses control instability in grid-forming inverters that use virtual admittance with voltage-feedback decoupling. Unity-gain decoupling removes necessary low-frequency stabilization, creating a delay-sensitive double-integrator structure. The authors introduce a proportional active-damping path via negative capacitor-voltage feedback that restores stability while preserving decoupling benefits, reducing voltage settling time from 9 ms to 3 ms in 3-kVA experiments.

Contribution: Identifies and solves the 'VFD trap' where full voltage-feedback decoupling destabilizes virtual-admittance control by removing capacitor restoring terms, proposing a compact active-damping solution with analytical design equations.

Relevance to your work: tangential

The Large Cancer Assistant (LCA): A Model-Agnostic Orchestration Framework for Scalable Clinical Decision Support in Oncology

Ghassen Marrakchi, Basarab Matei · arXiv cs.LG · 2026-07-07

★☆☆☆☆ This work addresses software architecture for clinical oncology decision support systems with no connection to wireless communications, physical layer design, or 6G technology development.

This paper proposes the Large Cancer Assistant (LCA), a model-agnostic orchestration framework for clinical decision support in oncology that decouples data ingestion, routing, and AI inference through a 7-tuple architecture enforcing 'Algorithmic Impermeability.' The framework uses Geometric Deep Learning to standardize multimodal patient data and outputs a Standardized Intermediate Payload (SIP) isolating AI execution from hospital IT infrastructure. A proof-of-concept validates negligible orchestration overhead, invariant routing under model swaps, 100% recall on data anomalies, and multi-protocol execution.

Contribution: The novel contribution is the formalization of a strictly model-agnostic orchestration layer for healthcare AI that maintains routing logic independent of black-box models, enabling seamless model swapping and EMR interoperability through architectural decoupling.

Relevance to your work: tangential

Towards Quantum Network Performance Metrics: Challenges and Demonstration

Mohamed Shaban, Mariam Kiran, Muhammad Ismail · arXiv cs.NI · 2026-07-06

★☆☆☆☆ While technically competent, quantum networking remains far from 6G terrestrial wireless deployment timelines and does not address RF physical layer, channel modeling, or spectrum-sharing challenges relevant to beyond-5G systems.

This paper proposes a performance monitoring framework for quantum networks, defining metrics across quality (entanglement fidelity, QBER), throughput/latency (entanglement rate), timing jitter, and environmental factors. The authors implement a non-invasive prototype at Oak Ridge National Laboratory to demonstrate real-time data collection and alert generation. The work addresses challenges in balancing observability with system performance and proposes foundations for autonomous quantum network control.

Contribution: First structured framework for quantum network performance metrics with a demonstrated real-world monitoring prototype, bridging quantum hardware operation with software-defined network management concepts.

Relevance to your work: tangential

MCP-Enabled Agentic AI for Autonomous IPoDWDM Network Lifecycle Automation

Chunmin Xia, Jakub Harbaczewski, Nikhil Dsilva, Julie Raulin, et al. · arXiv cs.NI · 2026-07-07

★☆☆☆☆ Focuses on optical transport layer automation rather than wireless air interface or 6G physical layer concerns; addresses infrastructure management outside the research scope of AI-native radio systems.

The paper demonstrates an agentic AI architecture using Model Context Protocol (MCP) for autonomous control of IP-over-DWDM optical networks. The system performs end-to-end lifecycle automation and closed-loop control using GNPY simulation tools and network telemetry. Validation was conducted on a physical testbed with vendor-agnostic multi-layer network equipment.

Contribution: First demonstration of MCP-based agentic AI framework for autonomous optical transport network control with live testbed validation across multiple vendors and network layers.

Relevance to your work: Tangential - while agentic AI and closed-loop control principles may inform autonomous network management in NTN or 6G RAN orchestration, the optical transport domain and IPoDWDM specifics do not directly connect to air interface research or beyond-5G wireless system design.

AI-native air interface

Bridging the Semantic Gap in 6G: Tiny Language Models Under the Latency-Accuracy-Size Trilemma

Arnav Mathur, Garima Mathur, Rahul Jashvantbhai Pandya · arXiv eess.SP · 2026-07-03

★★★★★ Addresses a critical practical bottleneck (compute/memory/latency constraints) in deploying AI-native semantic communication at 6G scale, with concrete compression roadmaps and 3GPP standardization considerations relevant to IMT-2030.

This review addresses the mismatch between large transformer-based semantic encoders and the stringent latency/memory/energy constraints of 6G edge devices by surveying tiny language models (t-LMs) and compression techniques. It provides a taxonomy of t-LM-based semantic communication architectures, analyzes quantization/pruning/distillation methods that preserve semantic quality, and reviews resource allocation frameworks for multi-user 6G networks. Empirical evidence shows 99.98% model size reduction with preserved task accuracy, split computing enabling 640-parameter on-device encoders, and 65% energy savings via knowledge graph integration.

Contribution: First comprehensive survey unifying semantic information theory, t-LM compression techniques, and 6G resource allocation, with a taxonomy spanning five architecture classes and quantitative benchmarks on extreme compression for IoT/edge semantic communication.

Relevance to your work: Directly relevant to your AI-native air interface work; provides design patterns and compression baselines for deploying learned semantic encoders on resource-constrained 6G endpoints, bridging theory and implementation.

Cognitive Digital Twins for Self-Aware Channel Estimation

Afan Ali, Ali Arshad Nasir, Daniel Benevides da Costa · arXiv eess.SP · 2026-07-05

★★★★★ Addresses a critical operational problem for deployed AI/ML PHY components and provides a 3GPP-compatible solution with demonstrated robustness across diverse channel conditions.

AI/ML channel estimators fail when real propagation deviates from training distributions. This work introduces a cognitive digital twin framework using a VAE with latent-space monitoring to detect distribution drift and trigger lifecycle actions (continue, update, retire) without needing ground-truth CSI. Simulations show the approach outperforms static deep learning estimators under moderate-to-severe channel drift.

Contribution: Model-agnostic VAE-based drift detection enabling autonomous lifecycle management of neural channel estimators, aligned with 3GPP AI-native framework requirements.

Relevance to your work: Directly relevant to AI-native air interface reliability and lifecycle management, especially for learned PHY and neural receivers that must adapt to non-stationary wireless environments in commercial 6G deployments.

Self-Supervised Implicit CEST Reconstruction via Physics-Informed Lorentz Encoding

Dexuan Li, Yupeng Wu, Chenglong Wang, Hanlin Liu, et al. · arXiv cs.LG · 2026-07-07

★☆☆☆☆ This is medical imaging methodology (MRI reconstruction) with no connection to wireless communication systems, channel modeling, or air interface design.

The paper addresses slow acquisition times in CEST MRI by proposing Lorentz Encoding, a physics-informed implicit neural representation that reconstructs high-resolution Z-spectra from sparse samples. Unlike generic coordinate encodings, LE constrains the continuous spectral mapping using parametric Lorentzian profiles as learnable basis functions, incorporating physical priors into self-supervised learning. On human brain data with 39-point sampling, LE achieves 57.58 dB PSNR and 0.9994 SSIM while preserving quantitative metabolite accuracy.

Contribution: The novelty lies in embedding physical Lorentzian models directly into the implicit neural representation's coordinate encoding, creating a self-supervised framework that respects spectral physics rather than relying on generic positional encodings.

Relevance to your work: tangential

ISAC (sensing + communication)

Rainbow Beamforming for Wideband LEO Satellite Communications: Principles, Applications, and Technical Challenges

Juha Park, Hyungseok Ko, Haejung Kim, Namyoon Lee, et al. · arXiv cs.IT · 2026-07-06

★★★★■ Strong conceptual contribution with direct NTN and ISAC relevance, from credible authors (Poor et al.), addressing real wideband LEO constraints for 6G non-terrestrial networks.

This paper reframes the beam-squint effect in wideband LEO satellite systems from an impairment to an exploitable resource. Rainbow beamforming deliberately uses frequency-dependent beam directions to create spatial diversity across subcarriers with minimal RF chains. The authors demonstrate applications in massive access (replacing beam hopping), integrated sensing and communication, and rapid satellite acquisition.

Contribution: Paradigm shift from mitigating beam-squint to intentionally exploiting it for frequency-spatial diversity in wideband LEO systems, enabling single-RF-chain multi-beam capabilities.

Relevance to your work: Directly relevant to your NTN and ISAC interests—rainbow beamforming offers a hardware-efficient approach to simultaneous sensing and multi-user communication in wideband LEO scenarios with frequency-selective channels.

Channel modeling / GAN-based

Rydberg Atomic Quantum Radio: A Comprehensive Survey From Wireless Communication Perspective

Yiyue Xiang, Neng Ye, Qihao Peng, Junrui Zhao, et al. · arXiv eess.SP · 2026-07-07

★★★★■ Well-motivated survey of an emerging quantum sensing technology with clear ISAC relevance, though RAQ receivers remain experimental with uncertain timelines for commercial 6G deployment at scale.

This survey reviews Rydberg atomic quantum (RAQ) radio receivers that map electromagnetic fields onto atomic quantum states, offering full-spectrum agility, ultra-high sensitivity, and anti-jamming capabilities beyond conventional RF front-ends. The paper bridges atomic physics and wireless communications by analyzing quantum mechanisms, architectural tradeoffs (sensitivity vs. bandwidth vs. frequency), and equivalent channel models for RAQ systems. Applications include cognitive radio, MIMO, satellite networks, ISAC, and RIS-assisted deployments relevant to 6G SAGSIN architectures.

Contribution: First comprehensive survey mapping RAQ radio physics to wireless channel models and communication-theoretic performance limits, systematically reviewing sensitivity-bandwidth-frequency tradeoffs and identifying RAQ deployment scenarios for next-generation integrated networks.

Relevance to your work: Directly relevant to ISAC research as RAQ radios enable simultaneous ultra-sensitive spectrum sensing and reception; channel modeling framework may inform future quantum-classical hybrid receiver architectures for 6G NTN and sub-THz bands.